

Deepak Mallampati

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PROFESSIONAL SUMMARY

Forward-deployed AI engineer with 2.5+ years building customer-facing AI agent integrations, multi-system automation workflows, and production LLM/RAG pipelines. Skilled in TypeScript, Python, REST/webhook API integration, and rapid prototyping of AI-powered customer experiences. Proven track record shipping production systems: ~35% retrieval precision improvement, >30% hallucination reduction via agent reasoning controls, and real-world multi-provider API orchestration. Strong technical advisor instincts - able to translate complex AI capabilities into customer outcomes and guide adoption of AI-native platforms.

CORE COMPETENCIES

AI Agent Development & Deployment • LLM / RAG / Function Calling • Prompt Engineering & Evals • TypeScript / JavaScript • Python • REST API Integration • Webhooks & Event-Driven Architecture • FastAPI • Docker • CI/CD • Semantic Search • Multi-Agent Orchestration • SQL & Data Analysis • Customer Technical Advising • Rapid Prototyping

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology & AI Consulting

- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation) with structured reasoning, tool discipline, and grounding rules; reduced hallucinations by **>30%** and improved response stability by **~25%** - directly applicable to building reliable customer-facing AI agents.
- Built **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer-based embeddings improved retrieval precision by **~35%**; response alignment exceeded **90%** in evaluation.
- Shipped **FastAPI/Flask** RESTful inference APIs containerized with **Docker**; structured logging and load simulation reduced post-deployment defects by **~30%**.
- Built data preprocessing pipelines (Pandas, NumPy) for EHR data; raised dataset reliability above **98%** via data quality validation and lineage documentation.
- Maintained HIPAA-conscious **AI governance**: audit trails, system-limitation docs, evaluation records - establishing AI observability and responsible deployment standards.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Enterprise ERP

- Optimized T-SQL stored procedures; reduced query execution time **~20%** and retrieval latency **~25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates; reduced deployment errors **>30%**.
- Designed **Grafana** dashboards for query performance and incident tracking; improved root-cause resolution speed **~30%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Applied AI

- Deployed **Flask-based API** with document Q&A (RAG); reduced hallucinated outputs **~30%**; raised contextual accuracy **>85%** for non-technical stakeholders.

SELECTED PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Translation shipped

Production **TypeScript** browser extension (Manifest V3, service workers) integrating **four AI providers** via REST APIs (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-section capture pipeline with coordinate remapping, 7-day cache, per-domain selector configs for 29 manga/webtoon sites - demonstrates production-grade multi-vendor API integration and orchestration complexity comparable to DevRev integration work.

TypeScript, REST APIs, Manifest V3, service workers, LLM orchestration, multi-provider integration

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Risk Scoring) with **schema-validated JSON contracts** between agents to prevent cascading errors. RAG-grounded CPT/ICD validation; ANN similarity search for duplicate detection; explainable risk scoring with audit-ready traces - parallels DevRev agent + integration automation patterns.

Python, LangChain, multi-agent orchestration, RAG, vector search, FastAPI, structured outputs

Dream Decoder - AI-Powered Multimodal Platform

Full-stack **FastAPI + React/TypeScript/Vite** app with coordinated multimodal APIs. Intermediate structured prompt transformation layers improved contextual alignment **~30%** and first-pass image success **~25-30%**.

FastAPI, React, TypeScript, Vite, Tailwind CSS, LLM APIs, prompt engineering

Geospatial Road Network Analysis

Parsed raw OpenStreetMap XML into a weighted NetworkX DiGraph; Dijkstra routing; betweenness centrality on a 20k-node subgraph; Louvain community detection; ~30% runtime reduction via subgraph scoping - demonstrates graph data structures and data mapping proficiency.

Python, NetworkX, graph algorithms, data mapping, schema alignment

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), FastAPI, Flask, React

AI & LLM: LLM application development, RAG, agentic workflows, prompt engineering, LLM evals, function calling, semantic search, embeddings, LangChain, LlamaIndex, PyTorch, Hugging Face

Integration & APIs: REST API design, webhooks, event-driven architecture, multi-provider orchestration, API integration patterns

Infra: Docker, Jenkins, CI/CD, Grafana, cloud-ready deployment, structured logging, observability

Work Authorization: F-1 OPT - authorized to work in the United States now